Designing with Impaired Technologies: A Game Jam of Constraints and Affordance Shrinkage

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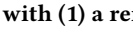
Figure 1: Top (left to right): Peripherals with visible impairments used as game jam input devices. Bottom: Participants playing with (1) a remapped keyboard, (2) a controller operated with a pen due to missing buttons, and (3) a half-damaged screen.

Abstract

Playing under constraint, whether due to disability, financial limitations, technological obsolescence, or other factors, is a widespread but under-examined aspect of gaming culture. We introduce "impaired technologies" (damaged peripherals such as cracked screens, controllers with missing buttons, and faulty speakers) as design provocations to explore what happens when the affordances available for play "shrink." Drawing on Dokumaci's concepts of "affordance shrinkage" and "activist affordances," we hosted a two-day public game jam where participants (n=9) first played existing games on impaired technologies, then designed their own games inspired by the experience. We identified six adaptive play strategies improvised under constraint, and three design themes, transforming "bugs" into features, embedding constraint within gameplay, and prioritizing fun over challenge. Our findings position constrained play as a site of creativity and agency, with implications for accessibility, jank, sub-optimal hardware gaming, and inclusive design.



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CCS Concepts

• **Human-centered computing** → *Interaction design*; *Interaction paradigms*; *Accessibility*.

Keywords

game jam, game design, play styles, adaptive, impairment, accessibility, inclusion

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1 Introduction

Games are now the world's most popular entertainment medium [41], yet poor accommodations and the lack of standardized accessibility guidelines exclude many players [28, 47, 59]. Accessibility is often an afterthought in game development, and it can be challenging to engage the game-playing audience with accessibility topics and inclusive experience considerations (e.g., low awareness [27, 33]). Inclusive design [57] means recognizing and designing for diversity in player populations, considering abilities, age groups, hardware and various other contextual factors and challenges. Prior research has focused on creating adaptations, such as via accessible game controllers [15], audio description [38], and fan mods [20]. Such accommodations can help, yet retrofitting accessibility in this way can often lead to tensions between accessibility and appropriate challenge, or oversimplification. We propose shifting the focus away from post-release accessibility to ask instead: *How can we design games that are inherently and broadly inclusive?*

To encourage alternative design paradigms for these inclusive games, we introduce the concept of "**impaired technologies**." These refer to damaged peripherals that might typically be discarded as e-waste, such as cracked screens, controllers with missing buttons, and unstable audio jacks. When the sensory interface for play is impaired, how do players adapt their behaviours and strategies? We draw a parallel between the challenges presented by these constrained experiences and the common but too rarely discussed experience of facing barriers to play, whether these barriers come from disabilities, economics or various other factors. We investigated the generative potential of impaired technologies through a two-day game jam, aiming to engage the public in our research while exploring the following questions:

- RQ1 What adaptive strategies do people use when playing games with impaired technologies?
- RQ2 What types of affordances emerge when people design games inspired by impaired technologies?

Addressing these questions, we analyze the adaptive strategies people develop when encountering impaired technologies, through the lens of Dokumaci's account of *activist affordances* [17]. These are affordances that were not designed or intended, but improvised or created "in the face of constraints, failures and losses" [17, p. 7] via the "activist" engagement of users. Dokumaci develops the concept through analysis of how people with disabilities respond to "shrinkage" in affordances, "when the environment's offerings narrow [...] into a set of constraints rather than opportunities" [17, p. 7]. However, she emphasizes its broader, intersectional applicability to any situation of affordance shrinkage, ranging from economic and resource constraints, to technological obsolescence and climate change. We introduce the concept to the field of games research, as we see its potential to shed light on a range of topics, from accessibility, to how players appropriate, exploit and enjoy "jank" in games [7], to the under-discussed but widespread practice of gaming on sub-optimal hardware.

Our goal is not to "simulate" experiences of disability or economic constraint; these can be trivializing and perpetuate stereotypes [13, 37, 44, 53, 56]. Rather, we aim to create shared experiences where people respond to the shrinkage in their available affordances for play (e.g., creating or playing a game together using a partially shattered screen). To clarify the distinction, disabilities are often highly individual and shaped by unique circumstances. In contrast, impaired technologies can provide publicly shared experiences without requiring people to wear or attach foreign devices to their bodies (e.g., blindfolds or earplugs) that focus solely on the limitations encountered by constraining experiences. When interacting with impaired technologies, participants retain the full capabilities of their bodies while remaining free to adopt any additional adaptations or supports they choose. Following Dokumaci [17], we address constraint and affordance shrinkage as an increasingly common aspect of modern life and play, but also a potential site of player agency and creativity, opening up an "improvisatory space of performance" that, to paraphrase Dokumaci, lets players imagine the same games otherwise [17, p. 7].

2 Background

Game jams and hackathons – events where participants develop games or applications within a fixed, typically short, time frame – have been proposed as methods for conducting research [5, 22, 25] and supporting education [4, 35, 42]. During these events, participants creatively engage with topics through collaborative and speculative approaches. Prior work has found that playful game jams can facilitate more innovation through democratized game making [25] and critical thinking [24], which can support research through design practices or community-based activism (e.g., feminist intervention [31], indigenous culture [34], elections [51]). We see this methodology as aligning productively with Dokumaci's account of activist affordances [17], which emphasizes participants as creators, positioned between experts and end-users while participating in improvisatory creativity under constraint. Game jams likewise tend to emphasize the creative nature of constraint [36], and often aim to harness the creativity of non-experts [35].

There is potential to bring these perspectives together to shed light on player agency and creativity under conditions of constraint, an issue that connects to various themes in game literature. We give three examples, but readers will certainly think of others: First *jank*, where technical glitches and bad design create good or bad, but often uniquely wonky experiences, which users often appropriate and reinterpret giving them a new life [7]. Second, the experience of playing on sub-optimal hardware due to financial or

other constraints remains underexplored in HCI games research, despite being widely documented in other contexts, particularly across the Global South [e.g., 8, 16, 43, 45]. Notably, there is evidence of how players improvise around such constraints: Messias et al. [43], describe the appropriation and modding of Pro Evolution Soccer 2 by Brazilian players playing on outdated PS2 hardware.

Finally, accessibility is not yet a standardized practice within the video game industry [40], and as such, many accessible games are the result of retro-fitting, minor adaptations, and afterthoughts [10, 48]. Indeed, accessibility features are often implemented in response to community outcry [11] and sustained advocacy [3]. As such, many players for whom play technologies are not well adapted develop their own strategies and hacks to enable play [6, 46]. On Dokumaci's account, these and a range of other common situations can be seen as cases of affordance shrinkage: simultaneously sites of challenges and creativity [17]. We are interested in the potential of the game jam format to explore such situations and, inspired by Dokumaci, to see what emerges when we view access and gaming under constraint as intersectional issues rather than marginal experiences restricted to particular, excluded groups.

3 Method

Our study was approved by the ethics committee of the lead author's institution. We organized a two-day weekend game jam in a university building, running from 9AM to 5PM each day. We recruited participants through social media, game mailing lists, and word of mouth, making it clear that the event was part of a research study. Participants completed a consent form prior to the study. Of 24 responses, n=9 attended and took part in the study. All participants had hobbyist or lower experience of game design and development, with six participants stating it was their first game jam. Seven participants reported having little to no knowledge of accessibility, while the other two rated their knowledge level as hobbyist. Two participants reported being neurodivergent. Participants brought their own laptops for development and were not monetarily compensated for their time, but food and drink were provided during both days of the event.

We gathered impaired technologies through donations prior to the event and paired with appropriate games. For example, pairing *Overcooked 2* [55] and *Split Fiction* [26], games that rely heavily on visual information, with damaged monitors to highlight the challenges presented by the peripheral. Throughout the event, we set up six games, running on different impaired technologies at stations around the room (see Table 1). We aimed to represent a variety of genres and playstyles (e.g., competitive and cooperative, multiplayer and single-player). An audio-only game, *A Blind Legend* [19], ran on standard technology, to illustrate a different kind of constraint: a game designed without visuals.

On the morning of Day 1, participants played these games, socialized, and formed teams. During this time, researchers video-recorded and observed participants' gameplay before interviewing them in-situ about their experiences. Participants were encouraged to engage with the games however they preferred, whether by observing or playing themselves. Games could be played in any way, including adjusting game settings, so long as they were played on the paired impaired technology. Photos of four of the impaired technologies are shown in Figure 1. An experienced game jam facilitator ran a short activity guiding team formation, resulting in three teams: Team A (P1, P2, P3), Team B (P4, P5, P6), and Team C (P7, P8, P9).

Beginning in the afternoon of Day 1, the focus shifted to teams creating their own games. With an introductory presentation, we prompted participants to break conventional norms, and aim to include rather than to "fix" what seems to break conventional norms. Participants were able to use whiteboards, large sheets of paper, and their own laptops for documenting ideas and prototyping their games. This continued until the end of Day 2, when the teams presented their games to the rest of the group. Throughout the event, we interviewed each team about their design ideas and progress. On both days, during lunch, we hosted Lunch & Learn talks with academic and industry speakers presenting their work on games and accessibility.

3.1 Data Collection and Analysis

Participants completed pre- and post-event questionnaires anonymously, addressing their experiences of accessibility and gaming as well as event feedback. We also conducted semi-structured interviews throughout the event to keep up to date with participants' changing perspectives and ideas. During play with the impaired technologies, we observed participants' strategies and asked about their experiences, for example, "How did you change the way you played the game?" and "What did you find challenging about playing the game with an impaired technology?" To understand participants' design processes, we also conducted two group interviews with each team per day: on day one, after team formation and prior to departure; on day two, after arrival and after final presentations. Throughout the event, we took photos and videos of brainstorming notes and whiteboard sketches, and of participants playing the games. We collected a total of 32 videos and 139 photos.

The photos, gameplay videos, and interview videos were transcribed and inductively coded, which allowed us to identify categories and patterns and develop themes organically from participants' responses. For RQ1, two researchers independently open-coded the adaptive play strategies observed during play with the impaired games. They discussed their codes to reach consensus and used qualitative inductive content analysis [21] to develop broad categories of adaptive play strategies (see Section 4.1). We found a total of six distinct categories of adaptive play strategies: game settings, trial and error, bodily manipulation, finding fun, collaboration, and giving up.

For RQ2, data on the overall game creation process was divided among three researchers, with overlapping team assignments: one coded Teams A and B, another coded Teams B and C, and a third coded Teams C and A. The researchers convened to discuss and reconcile coding thematically for each team (see Section 4.2). Then, the three researchers identified and organized patterns shared across teams into overarching themes (see Section 4.2.4). Finally, three researchers who facilitated the event analyzed the pre- and post-event questionnaires (see Section 4.3 and 5.3). They discussed participant perspectives and reflected on potential adaptations to future events.

Table 1: Six games paired with impaired technologies for participants to play as a design provocation.

Game	Impaired Technology	Genres	Platform	Players
Overcooked 2 [55]	One taped up monitor with multiple areas on the screen obscured	Casual, co-op	Xbox Series X with four controllers	4
Brawlhalla [9]	Four game controllers with different missing or locked buttons	Fighting, multiplayer	PC (impaired controllers)	4
Split Fiction [26]	One cracked monitor with left half of the screen displaying yellow lines	Adventure, co-op	PS5 with two controllers	2
SpeedRunners [18]	One keyboard with missing keys and keys placed at incorrect locations	Racing, competitive	PC (impaired keyboard)	2
Alien: Isolation [14]	One pair of crackly speakers that play audio unreliably	Survival horror, FPS	Laptop with no external mouse	1
A Blind Legend [19]	No impaired hardware, but an audio game designed for blind players with no graphics on screen	Action, adventure	PC with speakers	1

4 Findings

4.1 Adaptive Play Strategies

When playing the six provided games (see Table 1), participants had to adapt to the affordances of the impaired technologies. We grouped our observation notes and interview transcripts to identify six categories of adaptive play strategy.

Game settings. The simplest play adaptation involved navigating around the reduced affordances via existing game settings. Across games with impaired inputs (*Overcooked 2* and *SpeedRunners*) and the audio game *A Blind Legend*, participants modified in-game settings such as button mappings to help them play. While playing *A Blind Legend*, P8 expressed a desire for three times faster narration speed, which was not an available feature.

Trial and error. Impaired input devices (controllers and keyboard) also provoked participants to adopt exploratory *trial-and-error* approaches. For example, P1 and P2 tested each key in the *SpeedRunners* game settings menu to understand mappings for missing keyboard inputs. Others described their strategies for playing *Brawlhalla* using controllers with missing buttons: "Hey this is a strategy: press every button until something works!" (P8).

Bodily manipulation. Some participants improvised physical play strategies with external tools or their own bodies. Playing *Brawlhalla*, for example, P2 found and used a pen to activate buttons on an impaired controller (see Figure 1). Playing *Overcooked 2* and *Split Fiction* with obscured or cracked screens, P8 physically leaned in to look closer at the onscreen action, while P4 and P7 physically gestured to the screen to help each other locate obscured objects.

Finding fun. Several participants were prompted by the affordance shrinkage of the impaired technologies to create their own alternative forms of fun, beyond those intended by the designers. As P4, P5, and P6 played *Overcooked 2*, multiple fires broke out in areas of the kitchen that were obscured under tape. While this shrinkage of affordance prevented players from playing the game as intended, P4 laughed "The whole kitchen went on fire; it became a fire-fighting game instead of cooking!" P7 and P9 reported similar experiences in *Split Fiction*, "you can have different levels of fun with it. I'm gonna go ahead and win, and laugh at your characters repeatedly falling off a rainbow road."

Collaboration. In collaborative games, impaired technologies could provoke greater and more intentional coordination with other players. In *Split Fiction*, P7 described needing to "just slow down [...] because it's like, I like to succeed. And if my friend does not succeed, I cannot succeed, which would make me mad." They also found that they "had to use the other person's screen more to try to, uh, locate [themself]," as their own portion of the monitor was damaged.

Giving up. Not all participants were willing to play every game. One participant's (P9) behaviour did not fit in the other categories as they simply gave up on some games. Watching others play *Overcooked 2* with a taped screen, they said, "It's quite a mess. [...] I try to see other girls playing. [...] I just don't know what she's going to be doing." They were also confused by the lack of visuals on *A Blind Legend*, asking "What's that on the screen?" On being informed it was an audio game, they responded "ah" and walked away. Further work with more participants is required to determine whether this reaction is due to intrinsic factors (e.g., personality).

4.2 Affordances Throughout Design

The three teams were provided with the same prompt but created distinctly different games (see Figure 2). For each game, we summarize the details the team's implementations, affordances, and design processes in the following subsections.

4.2.1 Team A (P1, P2, P3) : Africa Great Migration. Team A created 4-player 2D platformer in which each player has only two directional controls (up/down or left/right) and must race the others to the finish line. They found that the missing input affordances of the broken controllers could be reconfigured as positive affordances for simple **simple, limited, and asymmetric player controls**. To manage the restricted controls and overcome obstacles, players must activate mechanisms on the map. The team aimed to create an asymmetrical shared experience. As described by P3, "Everybody know how to play this game, but it's very hard to find the answer how to [win] the game." They took inspiration from their shared experience playing *Brawlhalla*: "where some of the controls weren't working, and you have to make do with the controls that you have. So we're designing a map around the limited controls" (P2). Connecting to to our "Finding Fun" theme, above, P2 explicitly described trying to incorporate the positive confusion they experienced in play with impaired technologies: "It's kind of that feeling of, you know how to play the game [but] *something was missing or something was broken [...] It was confusing*, was the main feeling that I had on Saturday morning. So I think we want to try to incorporate that kind of confused feeling into the game, through controls."

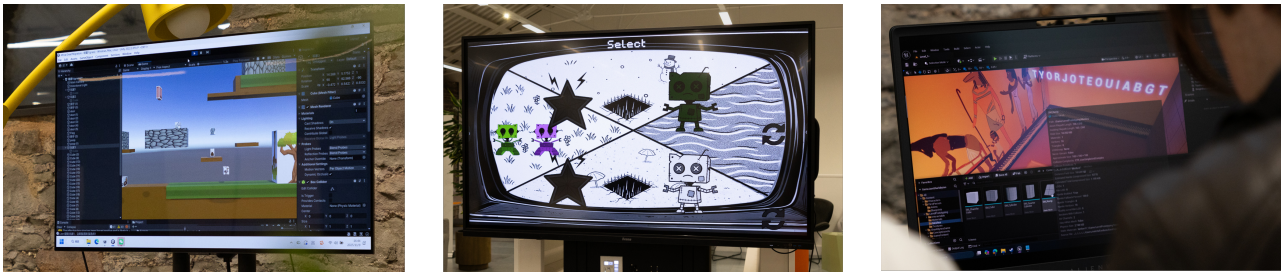

Figure 2: Photos of the games from left to right: Team A's *Africa Great Migration*, Team B's *Blackout*, and Team C's *Escape Tomb*.

4.2.2 Team B (P4, P5, P6): Blackout. Team B created a 4-player competitive game where players move across four zones to attack each other during blackouts, but do not see results until lights return. Sound and visual effects help players guess each other's locations in the darkness. They described how their game was built on the idea that experiences of impairment (using the same impaired screen) could vary across players and situations. P4 described playing *Split Fiction*: "at each moment the impairment to the screen affected us differently. And so sometimes actually it was very noticeable. Sometimes it was actually catastrophic and made it impossible to play the game, and other times it was just a mild inconvenience." Team B found that a shared experience of impairment afforded **shared uncertainty and inference of feedback during the screen blackout** before revealing outcomes. They described how they wanted to make use of impairments – implemented as complete darkness — but in a game playable without impaired technologies. P4 explained: "all of the gameplay actually happens when you can't see," citing inspiration from *Split Fiction*'s broken screen, where players "literally couldn't see half the map" and had to guess, look at another's screen or "hope for the best" P6 added, "the active bits that are being affected by the impairment and you're trying to overcome that."

4.2.3 Team C (P7, P8, P9): Escape Tomb. Team C created a single-player point-and-click puzzle game based on visual occlusion, inspired by playing with the taped screen. They explored mechanics in which the player would manipulate mummy bandages, adding or repositioning them across the screen, as a central method for solving puzzles in an escape room scenario set in an Egyptian tomb. Team C was inspired by their experience playing with the taped screen, finding that the missing visuals afforded **hiding or revealing on-screen elements for solving in-game puzzles**. This team prioritized making the game fun but challenging while limiting frustrations. P8 described how "an impairment can actually be used as benefit," looking beyond it as a source of difficulty in "basic parts of the game," but as a "positive element" P7 likened it to being thrown into an escape room with impaired vision, tasked with solving it "in the most fun way possible because there is no time limit."

4.2.4 Design Processes. We identified three broad common themes across the teams. **"Bug" to feature** captures when participants described impairments as "bugs," flaws, or constraints that were part of gameplay. As P4 described Team B's game concept, "we sort of designed the game to have [visual impairment] as a *main feature not a bug*." Likewise, P2 highlighted that Team A was "designing a map around the *limited controls*." **Implementing impairment** refers to how teams tried to implement these features. Due to the limitations of our venue, participants were unable to stay overnight to work on their games, hence they were not able to develop their games using an impaired device. As P5 said, "we're trying to let the players adapt to the impaired experience by the gameplay design, not the direct using the impaired device." Finally, teams tended to **prioritize fun over challenge,** by aiming to create enjoyable games that limited difficult experiences. While fun and challenge are not mutually exclusive, some participants interpreted certain features as potentially frustrating for players. P8 described Team C's goal was "*not making it too annoying*, like you're constantly having some bandages, but it's still enough where it's a challenge." Similarly, P2 highlighted keeping Team A's game straightforward, "It's *very simple controls*. We've had a lot of fun designing it, so I think it's definitely going to be a lot of *fun to play*."

4.3 Participant Feedback

Participants were highly positive about the game jam. All reported that they would participate again, and enjoyed playing games with impaired technologies. They felt the impairments provoked them to invent "different ways of playing in order to get around the problems," and "rethink how to play games," provoking divergent thinking on issues such as "how to move about spaces" and how to "help a partner play." One participant felt they learned "ways to think about impairments and accessibility in terms of mechanics rather than add-ons." Another saw opportunities to explore other forms of breakdown: "bugs in the code can be as interesting to explore as impaired devices."

5 Discussion

5.1 Impaired Technologies as a Design Provocation

Participants perceived playing games with impaired technologies as a fun challenge that inspired them to incorporate aspects of that experience into their game concepts. The "confusion" described by Team A, and subsequent hands-on exploration, led participants to adapt how they played these games. As "activist affordances" is a term coined, primarily for the disabled lived experience [17], we use "adaptive strategies" to explain how participants created new ways to engage with the games alongside the shrinkage of available affordances. For example, Dokumaci discusses pain as a motivator for bodily manipulation to put on a pair of trousers.

Our participants leaned and moved their bodies (or tools, such as a pen) to play the games. Through play, participants were creatively engaged. Team A (P1, P2, P3) was inspired by how "some of the controls weren't working" (P2) while playing Brawlhalla together. Rather than seeing it as a setback, they adapted their playstyle — an experience that inspired them to design a game built around intentionally limited controls.

More broadly, our design provocation encouraged participants to consider the impairment as part of the play experience. This connects our approach not only to activist affordances but to other creative currents that can be seen as addressing shrinkage in affordances. For example, Glitch feminism [50] invites individuals to embrace glitches and find the liberation within them, much like how individuals find queer liberation in jank games [29, 52]. Similar to activist affordances, Russell [50] writes that glitches catalyze movement and change while simultaneously blocking and preventing. For example, Garrett et al. [23] found that glitches became sites of ethical exploration, where misalignment of crip feminist values [2] became moments of growth and interdisciplinary connection for everyone on the team. Impaired technologies could encourage designers to consider how these alternative modes of interaction could be creatively transformed into engaging gameplay experiences for a broader audience.

5.2 "Impairment" in Play

The teams interpreted and conveyed "impairment" differently with their games. For Team A, impairment as a design constraint was perceived as needing to limit interactions for all players. By keeping their game concepts simple and intuitive, Team A tried to design for a wider range of people. In alignment with universal design approaches [54], their implementation reduces complex interactions for all players as a design choice. Furthermore, they tried to embed feelings they experienced as non-disabled people using impaired technologies within their game (e.g., fear, anxiety, confusion). Although these emotions could lead to feelings of pity [44] as criticized with disability simulations, they could likewise contribute to empathy and help to educate players of their game.

Team B interpreted impairment as not being static and that it may not be "impairing" at all in certain contexts. By making what is not fully visible to players a core concept, their game demonstrates how shared experiences can still lead to different playstyles and strategies. Moreover, it is interesting that their goal of making their game accessible to players *without* impaired technologies directly parallels how games need to be made accessible for players with disabilities. However, the analogy between human impairment and digital bugs can be problematic because it may lead to a tendency to try to "fix" identities, akin to medical or normalization models of disability [60]. Further work can reflect on how social models of disability can be effectively conveyed to wider audiences. Moreover, other types of impaired technologies (e.g., glitches) could also be explored in future implementations of similar events.

Team C interpreted impairment as an alternative perspective for puzzle-solving. They had concepts for more interactive visual puzzles, but were unable to implement all of them. By prioritizing fun over challenge to avoid being "too annoying" (P8), their game highlights positive aspects of puzzle-solving but does not fully reflect the negative experiences of visual impairment. However, this positive conceptualization of impairment frames it as an alternative way of being, rather than as a challenge to be overcome in the game. Prior research suggests that playing disability simulation games (e.g., colour blindness) can foster empathy towards people with disabilities and encourage accessible design practices [32]. Future exploration can examine whether positive portrayals of impairments in games can reduce implicit bias or prejudice toward people with disabilities, such as through character representations (e.g., [12, 39, 58]). As a limitation, despite our aims to create constrained experiences with jank and sub-optimal hardware, these might still be interpreted as recreating experiences of impairment.

5.3 Reflections for the Future

We identified several areas where our speculative design research method could be improved for future related work. Most participants had little to no experience of games or accessibility, but reported being able to gain a better understanding. However, their existing skills limited their ability to create functional prototypes of their games to demonstrate. Additionally, the length of the event (2 days) may have been a barrier to other people participating. Future work can explore how game jams could be shorter and more compact by focusing on developing game concepts and possibly incorporating vibe-coding [49] to support functional prototyping. The findings of this exploratory study are limited by the small sample size of nine participants. Future research with a larger sample could further investigate and refine the categories and themes identified in this initial analysis.

As a public game jam event rather than a structured workshop, one limitation was that researchers could not strictly enforce equal participation in all activities. Participants could choose to spend more time playing one game, which could reveal whether the design provocation captured their interest. Although participants enjoyed playing the games using impaired technologies at their own pace, with some staying for long durations on certain games, we could more explicitly encourage them to try all the games to gain a greater variety of experiences. Furthermore, we suggest considering the types of games included; single-player games can provide more immersive experiences, but multiplayer games can offer shared social experiences and player discussion.

Moreover, we suggest the opportunity to implement similar event formats with different audiences, such as in museums or schools. Introducing accessibility topics through gameplay with impaired technologies could provide an interactive experience that socially engages children. Further discussion topics could include shared perspectives and "what is fair" in asymmetrical gameplay. Children could also create their own impaired technology-based games in teams as part of design-based learning [30] or counterstorytelling [1] activities to apply their understanding within their educational curriculum. Within this pedagogical approach, impaired technologies as a provocation could support inclusive design thinking and raise accessibility awareness. Future work can explore how the concept of "impaired technologies" can be applied across different contexts to support the introduction of accessibility and inclusion topics.

6 Conclusion

Participants in our two-day game jam began by playing existing games with "impaired technologies," improvising a range of adaptive play strategies to accommodate constraints. This inspired them to design their own games incorporating constraints and affordance shrinkage. The participants' games implemented intentional impairments, transformed "bugs" into features, and prioritized fun over "challenge." We see such design provocations as opportunities to provoke awareness of constrained play and promote creativity, cooperation and solidarity, building toward inherently inclusive game design.

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References

[1] Ezequiel Aleman, Michael Brown, and Ethan Ruchotzke. 2025. Listen up!: game jams as spaces of pedagogical rupture, counterstorytelling and youth agency. *Learning, Media and Technology* (2025), 1–14. doi:10.1080/17439884.2025.2600478

[2] Kafer Alison. 2013. *Feminist, Queer, Crip.* Indiana University Press, Bloomington and Indianapolis. https://www.jstor.org/stable/j.ctt16gz79x

[3] Sky LaRell Anderson. 2024. Video Game Accessibility Defined Through Advocacy: How the Websites AbleGamers.Org and CanIPlayThat.Com Use the Word Accessibility. *Games and Culture* 19, 5 (July 2024), 571–586. doi:10.1177/15554120231170156

[4] Riikka Aurava, Mikko Meriläinen, Ville Kankainen, and Jaakko Stenros. 2021. Game jams in general formal education. *International Journal of Child-Computer Interaction* 28 (2021), 100274. doi:10.1016/j.ijcci.2021.100274

[5] A. Tece Bayrak. 2017. Jamming as a design approach. Power of jamming for creative iteration. *The Design Journal* 20, sup1 (july 2017), S3945–S3953. doi:10.1080/14606925.2017.1352897

[6] Jen Beeston, Christopher Power, Paul Cairns, and Mark Barlet. 2018. Characteristics and Motivations of Players with Disabilities in Digital Games. arXiv:1805.11352 [cs.HC] doi:10.48550/arXiv.1805.11352

[7] Dan Bennett and Elisa D. Mekler. 2023. Jank Accounts: We Should Study 'Broken' Games. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play* (Stratford, ON, Canada) *(CHI PLAY Companion '23)*. Association for Computing Machinery, New York, NY, USA, 216–218. doi:10.1145/3573382.3616045

[8] M. Binark, G. Bayraktutan, and F. Buçakçı. 2008. Exploring Turkish Digital Game Culture in Internet Cafes. In *Proceedings: 6th International Symposium Communication in the Millenium İçinde, İstanbul Üniversitesi, Anadolu Üniversitesi ve University of Texas-Austin*. 14–16.

[9] Blue Mammoth Games. 2017. Brawlhalla. https://store.steampowered.com/app/291550/Brawlhalla/

[10] Mark Brown and Sky LaRell Anderson. 2021. Designing for Disability: Evaluating the State of Accessibility Design in Video Games. *Games and Culture* 16, 6 (Sept. 2021), 702–718. doi:10.1177/1555412020971500

[11] Geoffrey Bunting. 2025. As the Game Industry Cuts Back, Accessibility Is Feeling the Impact. https://www.theverge.com/games/669419/gaming-industry-accessibility-cutbacks.

[12] Diane Carr. 2019. Methodology, representation, and games. *Games and Culture* 14, 7-8 (2019), 707–723. doi:10.1177/1555412017728641

[13] David J. Connor and Lynne Bejoian. 2007. Cripping School Curricula: 20 Ways to Re-Teach Disability. *Review of Disability Studies: An International Journal* 3, 3 (2007), 11. https://rdsjournal.org/index.php/journal/article/view/285

[14] Creative Assembly. 2014. Alien: Isolation. https://store.steampowered.com/app/214490/Alien_Isolation/

[15] Mat Dalgleish. 2023. Who can play? Rethinking video game controllers and accessibility. In *Disability and Video Games: Practices of En-/Disabling Modes of Digital Gaming*. Springer, 43–71.

[16] Bruno de Paula. 2021. "Emergent Countries Play, Too!": The Zeebo Console as a (Partial) Decolonial Project. *Contracampo* 40, 2 (Sept. 2021). doi:10.22409/contracampo.v40i2.50267

[17] Arseli Dokumaci. 2023. *Activist affordances: How disabled people improvise more habitable worlds*. Duke University Press, Durham, North Carolina, United States.

[18] DoubleDutch Games. 2016. SpeedRunners. https://store.steampowered.com/app/207140/SpeedRunners/

[19] Dowino. 2015. A Blind Legend. https://store.steampowered.com/app/437530/A_Blind_Legend/

[20] Daniel Drak and Ben Barry. 2025. Disability Dress in Video Games: Player Modding and Fashioning Disability Worlds in The Sims 4. *ZoneModa Journal* 15, 2 (2025), 71–85.

[21] Satu Elo and Helvi Kyngäs. 2008. The qualitative content analysis process. *Journal of Advanced Nursing* 62, 1 (2008), 107–115. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1365-2648.2007.04569.x doi:10.1111/j.1365-2648.2007.04569.x

[22] Jeanette Falk Olesen and Kim Halskov. 2020. 10 Years of Research With and On Hackathons. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 1073–1088. doi:10.1145/3357236.3395543

[23] Rachael Garrett, Kat Hawkins, Patrick Brundell, Simon Castle-Green, Paul Tennent, Feng Zhou, Airi Lampinen, Kristina Höök, and Steve Benford. 2025. In the Moment of Glitch: Engaging with Misalignments in Ethical Practice. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–18. doi:10.1145/3706598.3713632

[24] Chloé Germaine and Paul Wake. 2025. Curious Games: Game Making, Hacking and Jamming as Critical Practice. *Behavioral Sciences* 15, 10 (2025), 1415. doi:10.3390/bs15101415

[25] William Goddard, Richard Byrne, and Florian 'Floyd' Mueller. 2014. Playful Game Jams: Guidelines for Designed Outcomes. In *Proceedings of the 2014 Conference on Interactive Entertainment* (Newcastle, NSW, Australia) *(IE2014)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/2677758.2677778

[26] Hazelight Studios. 2025. Split Fiction. https://store.steampowered.com/app/2001120/Split_Fiction/

[27] Michael Heron. 2012. Inaccessible through oversight: the need for inclusive game design. *The Computer Games Journal* 1, 1 (2012), 29–38. doi:10.1007/BF03392326

[28] Alexander Horowitz. 2024. *Accessibility by Numbers: A Critical Review of Game Accessibility Guidelines*. Springer International Publishing, Cham, 259–294. doi:10.1007/978-3-031-34374-2_10

[29] Claire Jackson. 2023. Cyberpunk 2077, Broken And Jank, Helped Me Bid Farewell To Closeted Life. https://kotaku.com/cyberpunk-2077-johnny-silverhand-v-lgbtq-trans-come-out-1850514969 Section: Kotaku Game Diary.

[30] Fengfeng Ke. 2014. An implementation of design-based learning through creating educational computer games: A case study on mathematics learning during design and computing. *Computers & Education* 73 (2014), 26–39. doi:10.1016/j.compedu.2013.12.010

[31] Helen W Kennedy. 2018. Game jam as feminist methodology: The affective labors of intervention in the ludic economy. *Games and Culture* 13, 7 (2018), 708–727. doi:10.1177/1555412018764992

[32] Devorah Kletenik and Rachel F. Adler. 2022. Let's Play: Increasing Accessibility Awareness and Empathy Through Games. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education - Volume 1* (Providence, RI, USA) *(SIGCSE 2022)*. Association for Computing Machinery, New York, NY, USA, 182–188. doi:10.1145/3478431.3499277

[33] Jozef Kulik, Jen Beeston, and Paul Cairns. 2021. Grounded Theory of Accessible Game Development. In *Proceedings of the 16th International Conference on the Foundations of Digital Games* (Montreal, QC, Canada) *(FDG '21)*. Association for Computing Machinery, New York, NY, USA, Article 28, 9 pages. doi:10.1145/3472538.3472567

[34] Annakaisa Kultima and Outi Laiti. 2019. Sami game jam–learning, exploring, reflecting and sharing indigenous culture through game jamming. In *Proceedings of DiGRA 2019 Conference: Game, Play and the Emerging Ludo-Mix*. doi:10.26503/dl.v2019i1.1099

[35] Gorm Lai, Annakaisa Kultima, Foaad Khosmood, Johanna Pirker, Allan Fowler, Ilaria Vecchi, William Latham, and Frederic Fol Leymarie. 2021. Two Decades of Game Jams. In *Proceedings of the 6th Annual International Conference on Game Jams, Hackathons, and Game Creation Events (ICGJ '21)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3472688.3472689

[36] Gorm Lai and Ilaria Vecchi. 2025. Formal Constraints and Creativity: Connecting Game Jams, Dogma '95, the Demo Scene, OuBaPo, and Renga Poets. *Games and Culture* 20, 8 (Dec. 2025), 963–983. doi:10.1177/15554120241233865

[37] Priya Lalvani and Alicia A. Broderick. 2013. Institutionalized Ableism and the Misguided "Disability Awareness Day": Transformative Pedagogies for Teacher

Education. *Equity & Excellence in Education* 46, 4 (Oct. 2013), 468–483. doi:10.1080/10665684.2013.838484 _eprint: https://doi.org/10.1080/10665684.2013.838484.

[38] María Eugenia Larreina-Morales and Carme Mangiron. 2024. Audio description in video games? Persons with visual disabilities weigh in. *Universal Access in the information Society* 23, 2 (2024), 577–588.

[39] Krystina Madej. 2024. *Representation of disability in Children's video games*. Routledge.

[40] Jesse J Martinez, Jon E. Froehlich, and James Fogarty. 2024. Playing on Hard Mode: Accessibility, Difficulty and Joy in Video Game Adoption for Gamers with Disabilities. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 524, 17 pages. doi:10.1145/3613904.3642804

[41] Sophie McEvoy. 2025. The Gaming Industry in 2025 by the Numbers — An Overview from GamesIndustry. https://www.gamesindustry.biz/gamesindustrybiz-presents-the-year-in-numbers-2025-year-in-review

[42] Mikko Meriläinen, Riikka Aurava, Annakaisa Kultima, and Jaakko Stenros. 2020. Game jams for learning and teaching: A review. *International Journal of Game-Based Learning (IJGBL)* 10, 2 (2020), 54–71. doi:10.4018/IJGBL.2020040104

[43] Jose Messias, Diego Amaral, and Thaiane Oliveira. 2019. Playing beyond Precariousness: The Political Dimension of Brazilian Modding in Pro Evolution Soccer. *Video games and the Global South* (2019), 75–86.

[44] Michelle R. Nario-Redmond, Dobromir Gospodinov, and Angela Cobb. 2017. Crip for a day: The unintended negative consequences of disability simulations. *Rehabilitation Psychology* 62, 3 (2017), 324–333. doi:10.1037/rep0000127 Place: US.

[45] Benjamin Nicoll. 2025. *Minor Platforms in Videogame History*. Routledge.

[46] Eunbyul Park, Jihun Chae, Karam Eum, Eunhye Choi, Hyunyoung Oh, and Young Yim Doh. 2025. Press Start to Continue: A Thematic Analysis of the Iterative Process of Hardcore Players with Disabilities Adapting to Gameplay Difficulties. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, 1–7. doi:10.1145/3706599.3719723

[47] Hwa-Jin Park and Sang-Beom Kim. 2013. Guidelines of serious game accessibility for the disabled. In *2013 International Conference on Information Science and Applications (ICISA)*. IEEE, 1–3. doi:10.1109/ICISA.2013.6579380

[48] John R. Porter and Julie A. Kientz. 2013. An Empirical Study of Issues and Barriers to Mainstream Video Game Accessibility. In *Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '13)*. Association for Computing Machinery, New York, NY, USA, 1–8. doi:10.1145/2513383.2513444

[49] Marco Riggio. 2026. *Cheap Fun: Accelerating Game Pre-Production via LLM-Based Vibe Coding*. Ph. D. Dissertation. Politecnico di Torino.

[50] Legacy Russell. 2020. *Glitch Feminism*. Verso, New York, NY, USA. https://www.versobooks.com/en-gb/products/460-glitch-feminism

[51] Anastasia Salter and Bridget M. Blodgett. 2017. This is fine: #ResistJam and the 2016 election in gaming. In *Proceedings of the 12th International Conference on the Foundations of Digital Games* (Hyannis, Massachusetts) *(FDG '17)*. Association for Computing Machinery, New York, NY, USA, Article 19, 10 pages. doi:10.1145/3102071.3102101

[52] M. D. Schmalzer. 2020. Janky Controls and Embodied Play: Disrupting the Cybernetic Gameplay Circuit. *Game Studies* 20, 3 (Sept. 2020). https://gamestudies.org/2003/articles/schmalzer

[53] Arielle M. Silverman, Jason D. Gwinn, and Leaf Van Boven. 2014. Stumbling in Their Shoes. *Social Psychological and Personality Science* (Nov. 2014), 464 – 471. doi:10.1177/1948550614559650 Publisher: SAGE PublicationsSage CA: Los Angeles, CA.

[54] Molly Follette Story. 2001. Principles of universal design. *Universal design handbook* 2 (2001), 4–12.

[55] Team17. 2018. Overcooked 2. https://store.steampowered.com/app/728880/Overcooked_2/

[56] Jan W. Valle and David J. Connor. 2019. *Rethinking Disability: A Disability Studies Approach to Inclusive Practices* (2 ed.). Routledge, London, UK. https://www.routledge.com/Rethinking-Disability-A-Disability-Studies-Approach-to-Inclusive-Practices/Valle-Connor/p/book/9781138085862

[57] Sam Waller, Mike Bradley, Ian Hosking, and P. John Clarkson. 2015. Making the case for inclusive design. *Applied Ergonomics* 46 (2015), 297–303. doi:10.1016/j.apergo.2013.03.012

[58] Tracy R Worrell. 2018. *Disability in the media: Examining stigma and identity*. Bloomsbury Publishing PLC.

[59] Bei Yuan, Eelke Folmer, and Frederick C Harris Jr. 2011. Game accessibility: a survey. *Universal Access in the information Society* 10, 1 (2011), 81–100. doi:10.1007/s10209-010-0189-5

[60] Zosia Zaks. 2024. Changing the medical model of disability to the normalization model of disability: Clarifying the past to create a new future direction. *Disability & Society* 39, 12 (2024), 3233–3260. doi:10.1080/09687599.2023.2255926